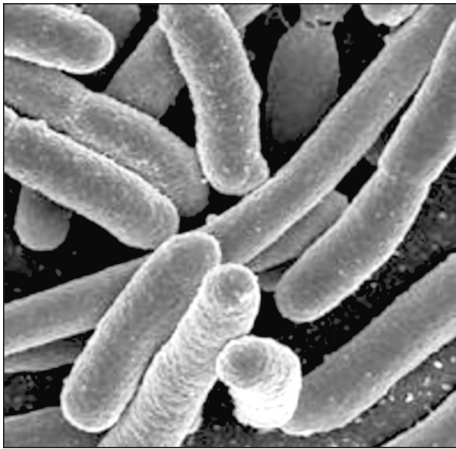


5. (a) The photograph shows bacteria called *E. coli*.



Bacteria such as *E. coli* can be pathogens.

(i) State the meaning of the term *pathogen*. [1]

(ii) Describe **two** ways in which pathogens are spread from person to person. [2]

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(b) Bacterial infections may be treated by antibiotics.

(i) State the name of **one** antibiotic. [1]

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(ii) Which **one** of the following results from the over use of antibiotics?
Underline the correct answer. [1]

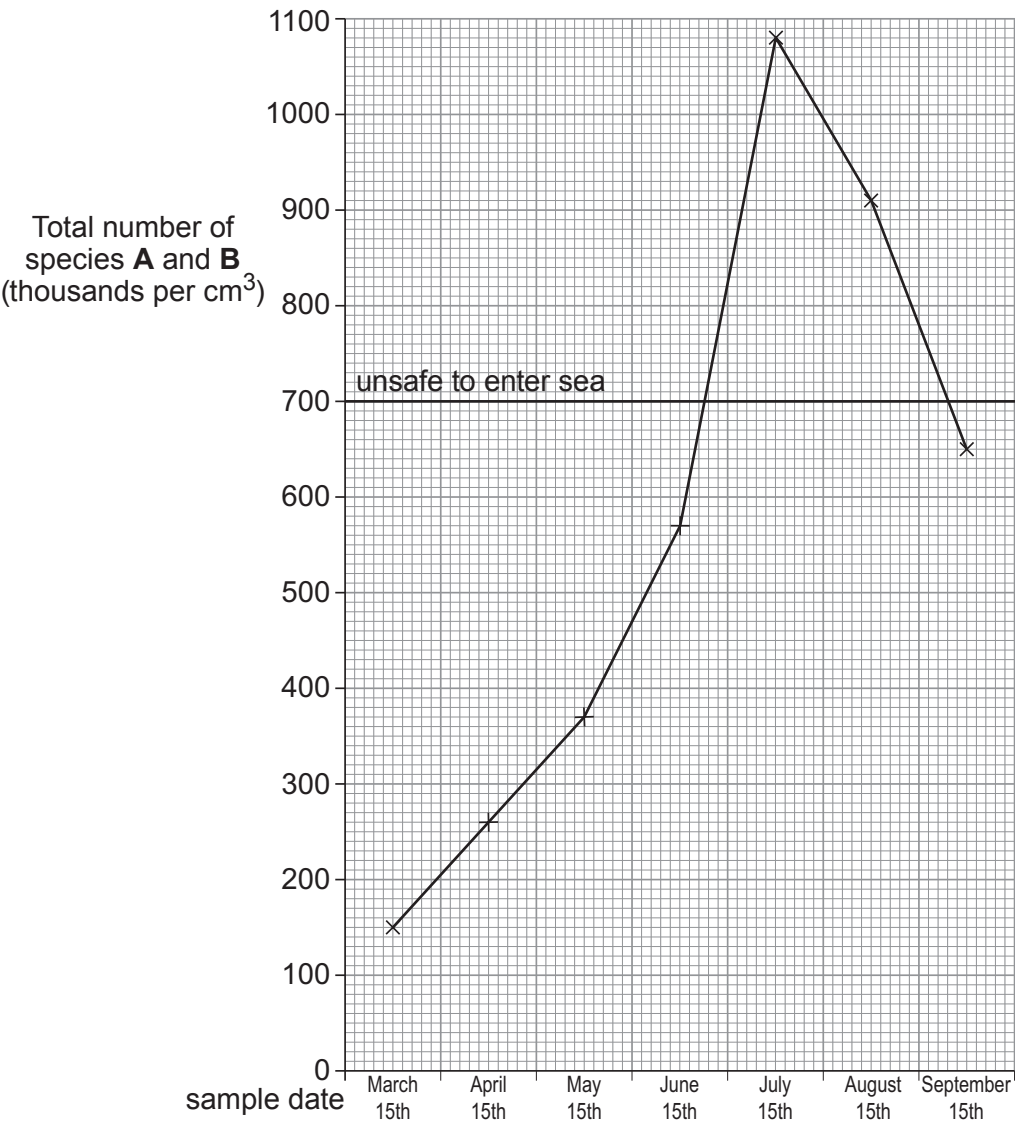
- People become immune to antibiotics
- People become resistant to antibiotics
- Bacteria become immune to antibiotics
- Bacteria become resistant to antibiotics



(c) The table shows the number of two species of bacteria (**A** and **B**) in samples of seawater taken near a beach.

Date of sample	Number of bacteria (thousands per cm ³)		Total number of species A and B (thousands per cm ³)
	Species A	Species B	
March 15 th	100	50	150
April 15 th	200	60	260
May 15 th	300	70	370
June 15 th	500	70	570
July 15 th	1000	80	1080
August 15 th	800	110	910
September 15 th	300	350	650

(i) The graph below shows how the total number of species **A** and **B** changed between March 15th and September 15th.



Examiner
only

- I. Between which **two** sampling dates was the increase in the total number of species **A** and **B** the fastest? [1]

Between and

- II. If the total number of species **A** and **B** in the sample is likely to be greater than 700 000 per cm³ in any month, people are advised to stay out of the sea.

A safety officer said:

“On June 15th, I decided to advise people to stay out of the sea until further notice.”

Explain why the safety officer made this decision, even though the seawater was safe on that date. [2]

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- (ii) I. Describe the trends shown in the **table** for the numbers of species **A** and **B**. [2]

Species **A**:

Species **B**:

- II. Use your answer to part (I.) to explain why the sampling should continue into October. [1]

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